Module 1: ggplot2 Practice

Lesson 4 | Visualize the textbook CSVs

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Preparation

Read Lesson 4. Use the checked CSVs in `data/module02/`. The data guide explains their variables and units. These exercises focus on constructing clear plots; Module 2 develops their statistical interpretation.

Tasks

1. Load Restaurant. Create a count bar chart for its quality ratings, ordered Good, Very Good, Excellent. Explain why `geom_bar()` is appropriate for raw rows.
2. Summarize those counts into a separate table, then reproduce the chart with `geom_col()`. Explain the difference between the two functions.
3. Load Audit. Draw a histogram of `audit_days` with bin width 5 and lower boundary 10. Label both axes and state what a bar represents.
4. Load Electronics. Draw a scatterplot of `commercials` against `sales_hundreds_usd`. Label the sales units explicitly. Add week labels without connecting the points.
5. Convert sales to USD in a new column and redraw the plot with corrected labels. Explain what changes and what remains the same visually.
6. Improve one chart using an informative title, a readable theme, and purposeful color. Explain why a continuous color gradient would be unsuitable for unordered brand categories.
7. Save one plot manually under `_output/` using `ggsave()`. Keep source data unchanged.

Avoid causal claims based on a chart alone. Use the shared course stylesheet for the report title; choose chart colors to communicate the data.

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